

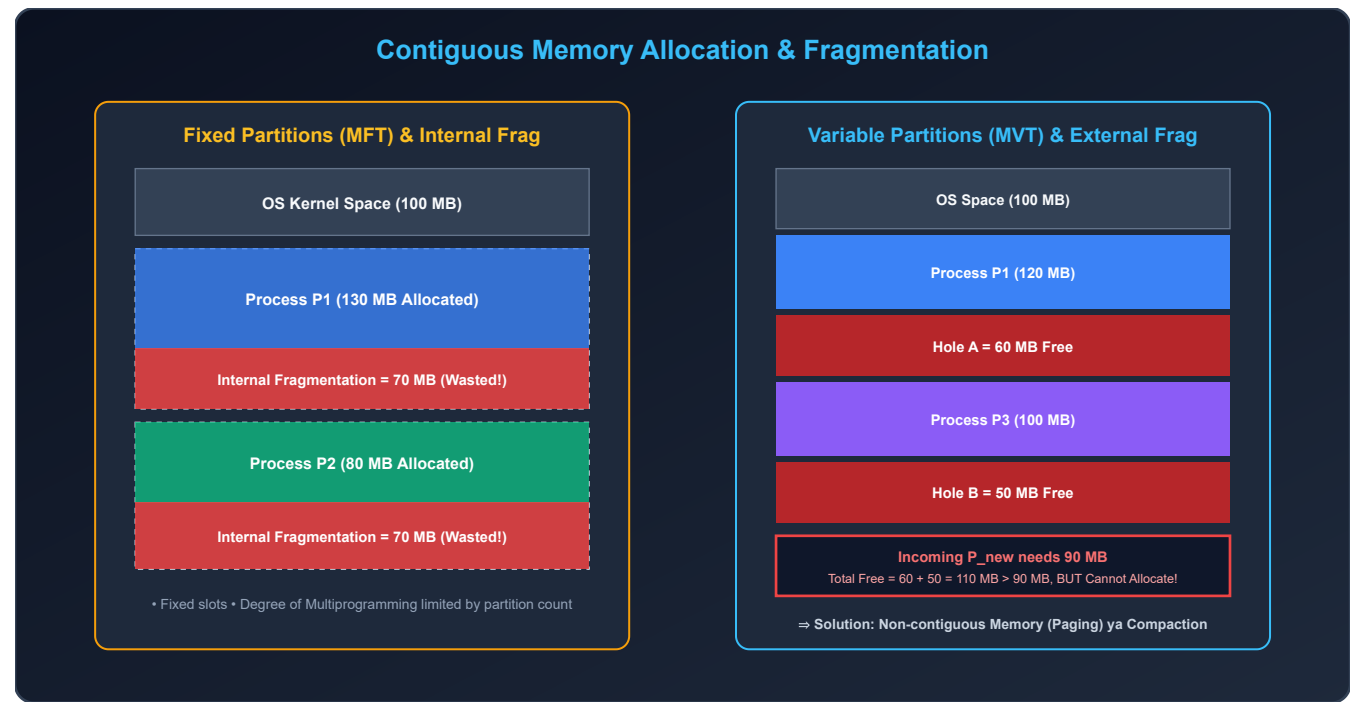
Module 02: Contiguous Allocation and Fragmentation

1. Contiguous Allocation Concept

Contiguous Allocation me har process ko RAM me ek unbroken continuous block of memory addresses assign hota hai.

2. Fixed Partition (MFT) vs Variable Partition (MVT)

Parameter	Fixed Partitions (MFT)	Variable Partitions (MVT)
Partition Size	Static / Boot time par predefined	Dynamic / Process requirement ke mutabik
Degree of Multiprogramming	Fixed (Number of partitions ke barabar)	Dynamic (Jitne process fit ho sakein)
Primary Fragmentation	Internal Fragmentation	External Fragmentation
Hardware Overhead	Very Low	Hole management table overhead



3. Internal vs External Fragmentation

Internal Fragmentation:

Partition process se bada hota hai, partition ke andar unused memory waste hoti hai.

External Fragmentation:

Total free memory sufficient hai, lekin scattered hone ki wajah se process allocate nahi ho pata (50% rule applies).

4. Hardware Protection: Base & Limit Registers

CPU logical address L generate karta hai. Agar $L < \text{Limit}$, tab physical address = $L + \text{Base}$. Otherwise, MMU ek trap generate karta hai (Segmentation fault).